Initial ionization from NEXUS tracks

NAUSICAA analysis

1 Calculation and interpretation

For each deposited-energy hit, the expected number of positive ions is

$$\overline{n}_i = 10^6 E_i[\mathrm{MeV}]/W[\mathrm{eV}], \qquad E_{\mathrm{evt}} = \sum_i E_i, \qquad \overline{N}_{\mathrm{evt}} = \sum_i \overline{n}_i.$$

We adopt $W = 21.9$ eV as a configurable nominal mean energy per electron–ion pair in gaseous xenon. It describes the partition of deposited energy between ionization and other processes, rather than the atomic ionization threshold; see Ref. [1]. This is an adopted input, not a measurement from these samples. Counts are continuous expectations, not rounded integer realizations. Each molecular Xe_2^+ ion carries charge $+e$.

The per-event mean ions per hit is $\overline{N}_{\mathrm{evt}}/N_{\mathrm{hits}}$. The table reports the mean and population dispersion of this quantity across events. The pooled value instead divides total ions by total hits. Hit-level quantities depend on the Geant4 deposition subdivision.

Each event is translated to the common generation vertex of its primary particles, copied from the original NEXUS particle table: $\mathbf{r}'_i = \mathbf{r}_i - \mathbf{r}_{\mathrm{vertex}}$. Orientations are preserved; no energy barycenter is used to define the origin. XY bins have pitch 2 mm. With the configured uniform speed 0.1 mm/ms, a 20 ms interval corresponds to a Z bin of 2 mm. The time axis is relative to the generation vertex, not an absolute arrival-time prediction. Projections are averaged over all saved events, including zero contributions, without normalizing individual event energies.

This stage describes initial ion production. It includes no ionization fluctuations, recombination, diffusion, focusing or collection losses. The two marginal projections do not determine joint pad-and-frame occupancy.

The spatial selection keeps $|x'| \leq 150.0$ mm and the same bound on $|y'|$. It is applied after translating to the generation vertex. No Z cut or subsequent recentering is applied. Raw energy totals include every original hit; the mean maps use the selected hits and the original number of events. The full uncut marginal histograms and per-event discarded energy are retained. Common grids enclose every selected hit with the configured margin; bins are left-closed/right-open, with the final upper edge included. Common XY grid: 168 by 168 bins, from -168 to 168 mm on each axis. Z: 1896 bins, from -1896 to 1896 mm.

2 Double beta

Saved events: 8995; hits: 2.833×10^6. Total expected ions: $1.008\,87 \times 10^9$. Pooled ions/hit: 356.115.

Quantity	Mean	Population SD	Minimum	Maximum
Energy (MeV)	2.456 29	0.013 206 8	2.300 09	2.457 83
Expected ions/event	112 160	603.049	105 027	112 230
Mean ions/hit in an event	360.986	41.3303	211.754	505.539
Hits/event	314.953	37.6124	222	530

The table above uses all original hits. After the vertex-relative XY cut, the mean retained energy is 2.418 07 MeV and the mean retained ion count is 110 414 per event. Discarded hits: 84 533 (2.983 87%). Discarded ions: $1.569\,93 \times 10^7$ (1.556 12%). Empty events: 0. Maximum discarded ion fraction in an individual event: 70.9945%. Discarded energy is never restored by rescaling.

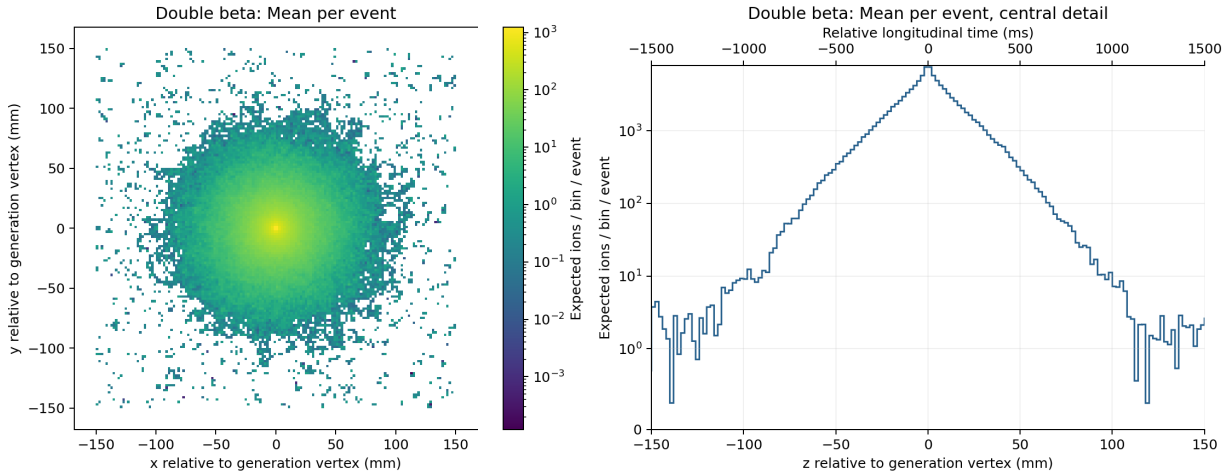


Figure 1: Selected ensemble mean, with zero contributions included in the denominator. XY is integrated over z; Z is integrated over x and y. The right panel shows the central longitudinal detail; it applies no additional Z cut.

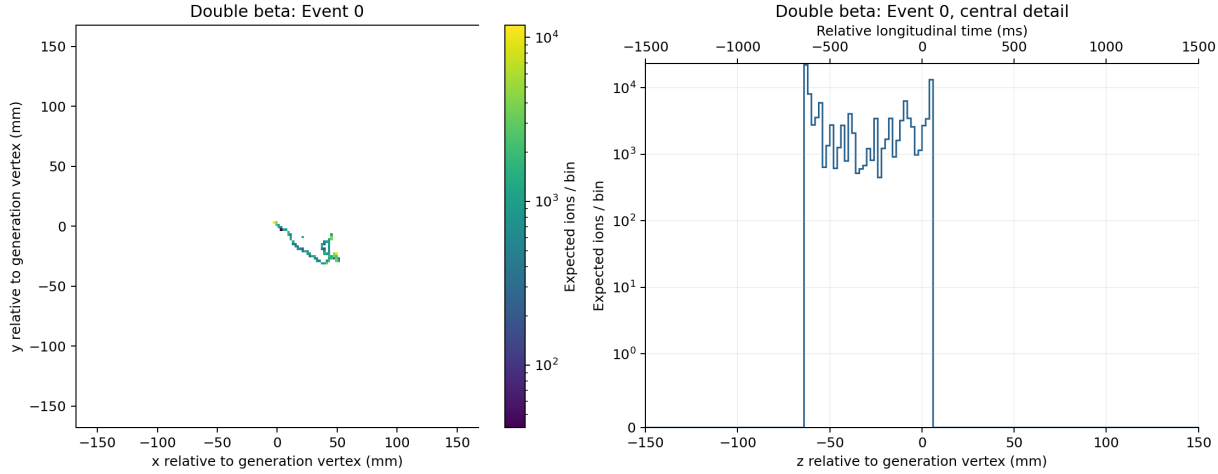


Figure 2: Selected hits of example event 0. XY is integrated over z; Z is integrated over x and y. The right panel shows the central longitudinal detail; it applies no additional Z cut.

Tail diagnostics

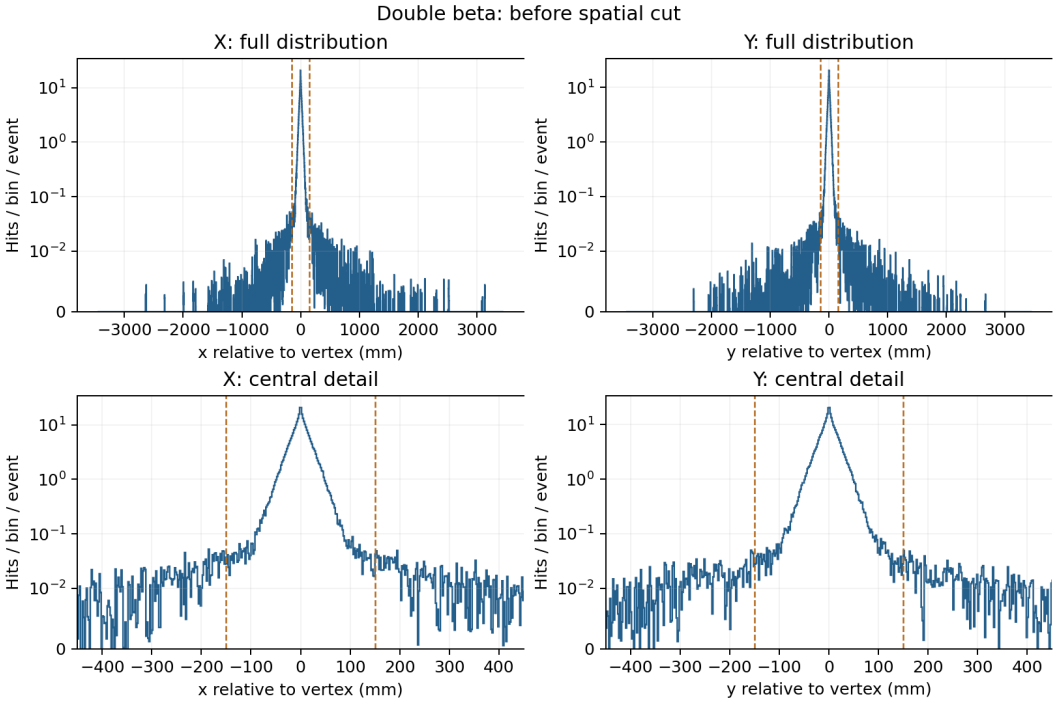


Figure 3: Uncut vertex-relative distributions. Dashed lines mark the XY cut; bottom panels show a detail of the same histograms.

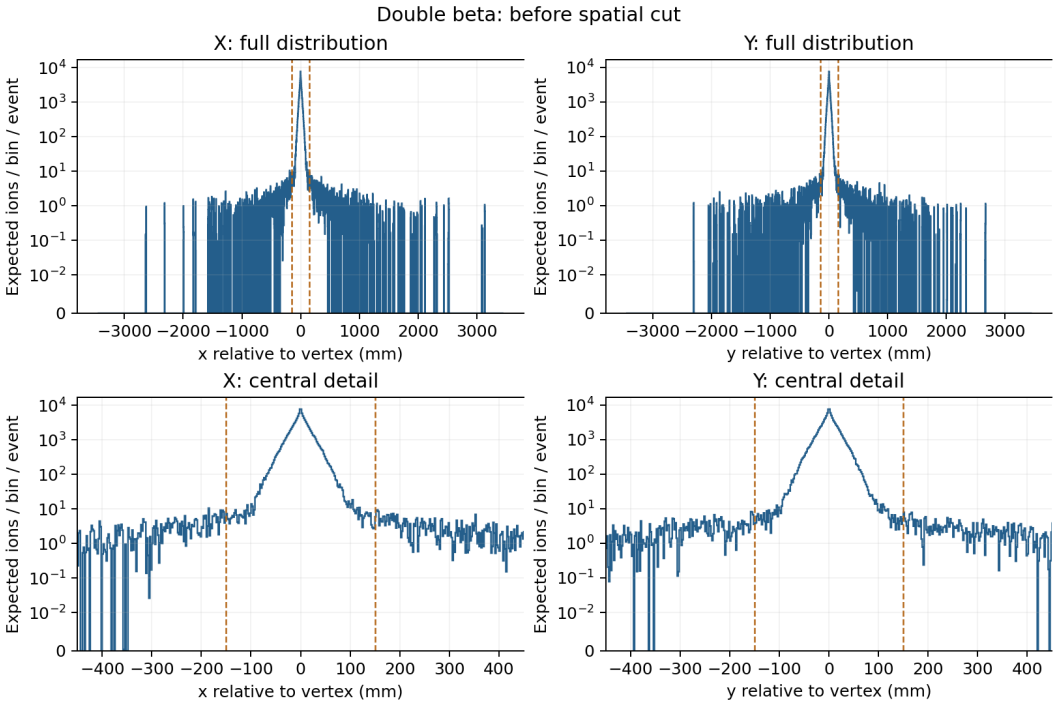


Figure 4: Uncut vertex-relative distributions. Dashed lines mark the XY cut; bottom panels show a detail of the same histograms.

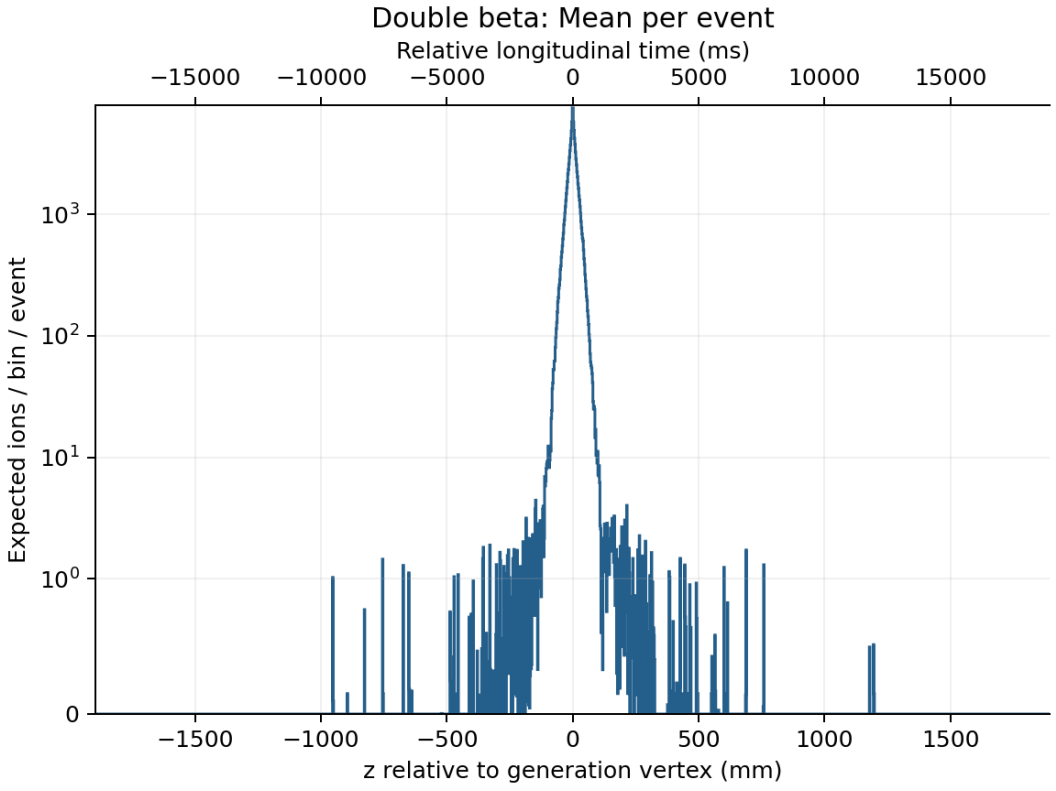


Figure 5: Full longitudinal distribution of retained hits.

3 Single electron

Saved events: 8483; hits: 2.7324×10^6. Total expected ions: $9.514\,37 \times 10^8$. Pooled ions/hit: 348.205.

Quantity	Mean	Population SD	Minimum	Maximum
Energy (MeV)	2.456 26	0.012 788 5	2.301 24	2.457 83
Expected ions/event	112 158	583.948	105 079	112 230
Mean ions/hit in an event	353.119	41.0734	211.355	503.272
Hits/event	322.103	39.048	223	531

The table above uses all original hits. After the vertex-relative XY cut, the mean retained energy is 2.394 25 MeV and the mean retained ion count is 109 326 per event. Discarded hits: 121 374 (4.442 03%). Discarded ions: $2.402\,03 \times 10^7$ (2.524 63%). Empty events: 0. Maximum discarded ion fraction in an individual event: 89.5198%. Discarded energy is never restored by rescaling.

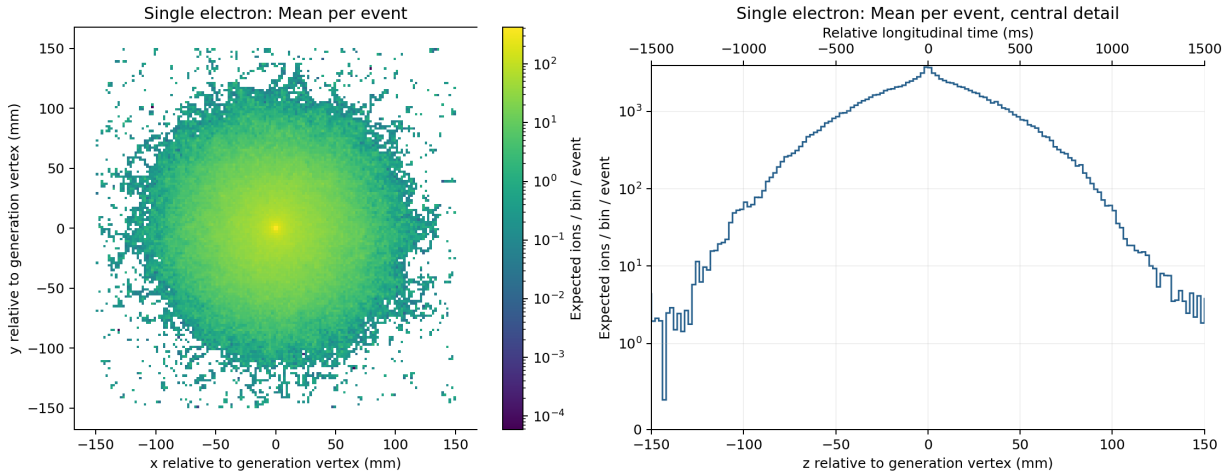


Figure 6: Selected ensemble mean, with zero contributions included in the denominator. XY is integrated over z; Z is integrated over x and y. The right panel shows the central longitudinal detail; it applies no additional Z cut.

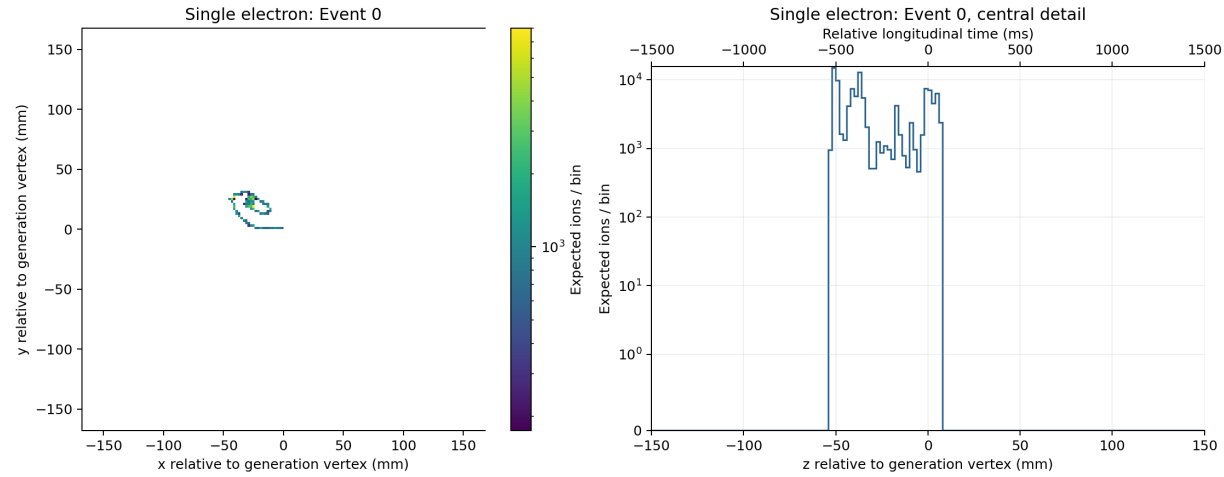


Figure 7: Selected hits of example event 0. XY is integrated over z; Z is integrated over x and y. The right panel shows the central longitudinal detail; it applies no additional Z cut.

Tail diagnostics

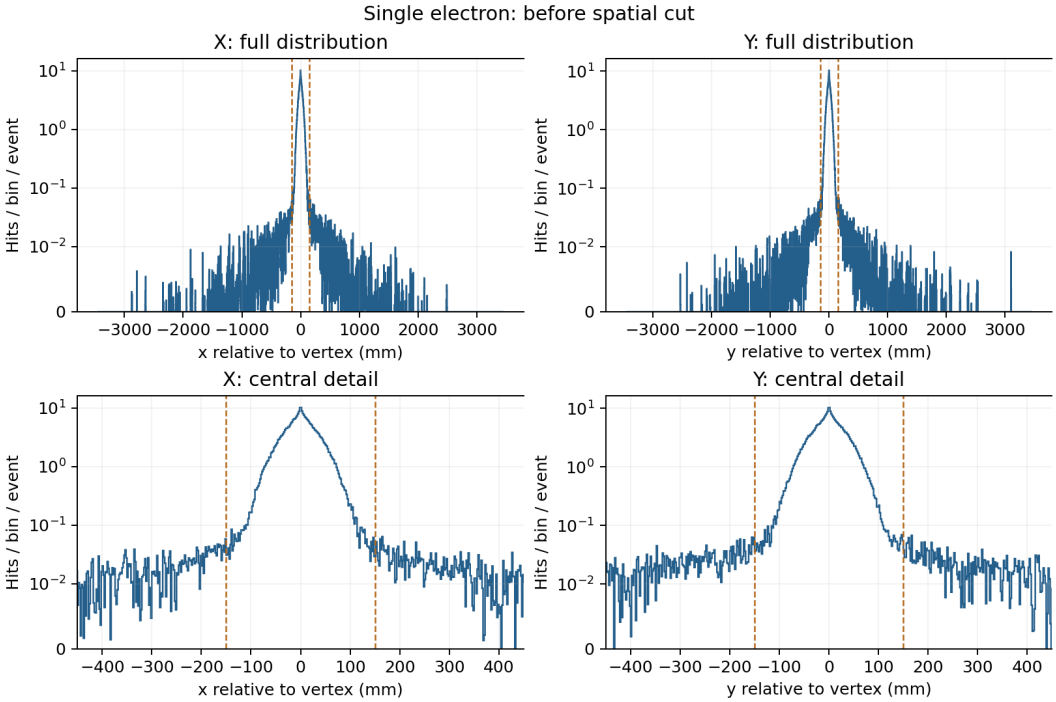


Figure 8: Uncut vertex-relative distributions. Dashed lines mark the XY cut; bottom panels show a detail of the same histograms.

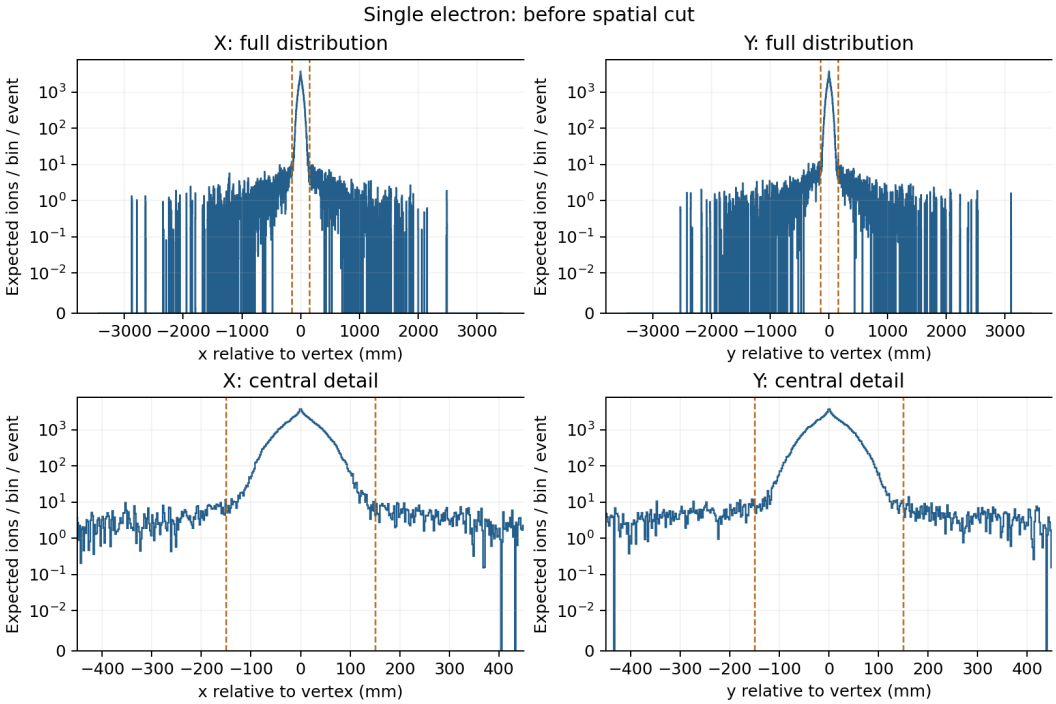


Figure 9: Uncut vertex-relative distributions. Dashed lines mark the XY cut; bottom panels show a detail of the same histograms.

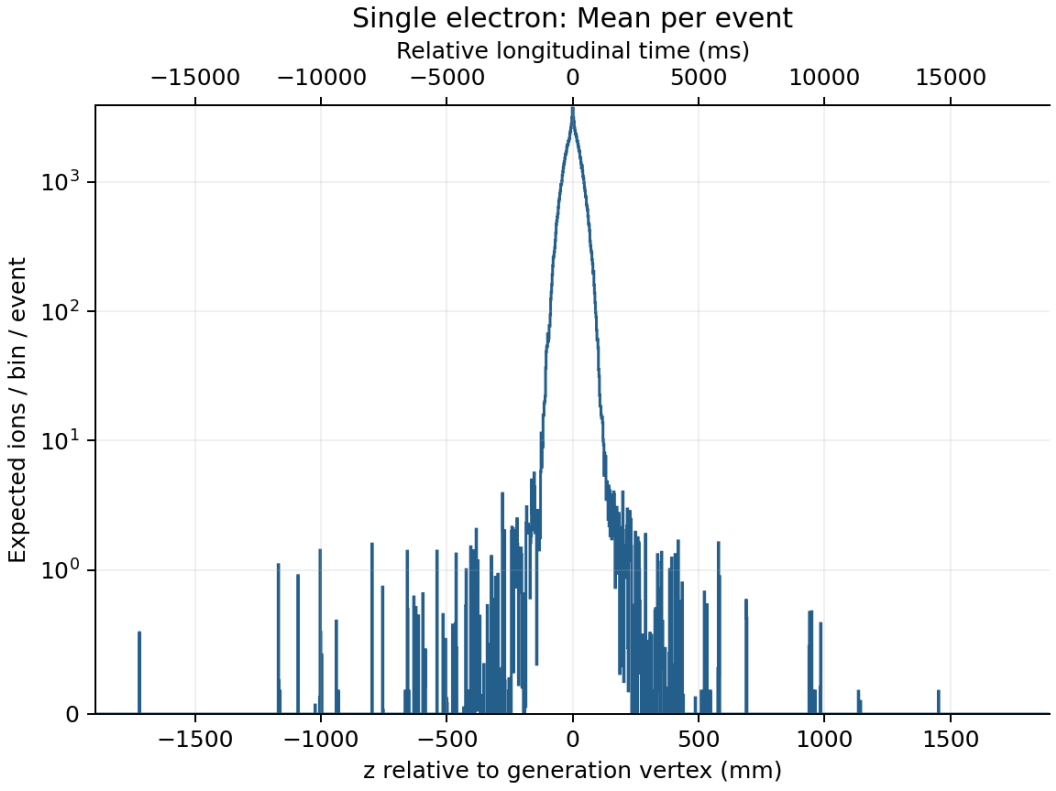


Figure 10: Full longitudinal distribution of retained hits.

4 Validation and provenance

The Julia calculation checks that each event's XY and Z projections sum to its retained expected ion count. The summed mean projections must equal the retained mean ion count per event. No renormalization is applied. Out-of-range hits cause an error. The following relative errors must be below 1×10^{-11}.

Sample	Worst event	Mean XY	Mean Z
Double beta	8.88×10^{-16}	2.22×10^{-16}	6.66×10^{-16}
Single electron	8.88×10^{-16}	1.11×10^{-16}	1.11×10^{-16}

Double beta

Input	events/0nubb_hitdf.h5
Original	0nubb.next.h5
Generator	Decay0Interface
Generated events	10000
Source lower energy cut	2300 keV
Source upper energy cut	2470 keV
Source radius	2. m
Source height	4. m
Source pressure	15. bar

Input SHA-256: `627a22476784b5279e8d716b581237c50789ba07ba1c80d0a573abfcb8c6e12f`.

Analysis code SHA-256: `6ed06e3d0e26540893703c8659bba7f37a569929b1cceadec1b66ad86e2b2573`. ormalsize

Single electron

Input	events/electron_0nubb_energy_hitdf.h5
Original	electron_0nubb_energy.next.h5
Generator	SingleParticleGenerator
Generated events	10000
Source lower energy cut	2300 keV
Source upper energy cut	2470 keV
Source radius	2. m
Source height	4. m
Source pressure	15. bar

Input SHA-256: `8d36747bba5b67515a9bf19d3d94aeb7252b3bf77f2a6583a1564c8a6852ecbe`.

Analysis code SHA-256: `6ed06e3d0e26540893703c8659bba7f37a569929b1cceadec1b66ad86e2b2573`. ormalsize

Numerical quantities, plots and tables are read from the Julia HDF5 result. The HDF5 files also retain configuration and software hashes, per-event quantities, bin edges and per-hit ion expectations in source row order.

References

[1] L. Serra et al. An improved measurement of electron-ion recombination in high-pressure xenon gas. *JINST*, 10(03):P03025, 2015.